W3a - 1.3 Max or Min of a Quadratic Function

MCR3U

1) Determine the vertex for each quadratic function by completing the square. State if the vertex is a maximum or a minimum.

a) $f(x) = x^2 + 14x - 14$

b) $f(x) = x^2 - 6x + 17$

c) $f(x) = x^2 + 7x + 11$

d) $f(x) = 2x^2 + 12x + 16$

e) $f(x) = -3x^2 + 6x + 1$

f) $f(x) = -\frac{1}{2}x^2 - x + \frac{3}{2}$

2) Use partial factoring to determine the vertex of each function. State if the vertex is a max or min.

a) $f(x) = 3x^2 - 6x + 11$

b) $f(x) = -2x^2 + 8x - 3$

c) $h(x) = -x^2 + 2x + 4$

d) $f(x) = 2x^2 + 12x + 17$

e) $f(x) = 4x^2 + 64x + 156$

f) $f(x) = \frac{1}{2}x^2 - 3x + 8$

3) An electronics store sells an average of 60 entertainment systems per month at an average of $800 more than the cost price. For every $20 increase in the selling price, the store sells one fewer system. What amount over the cost price will maximize revenue?

4) Last year, a banquet hall charged $30 per person, and 60 people attended the hockey banquet dinner. This year, the hall's manager has said that for every 10 extra people that attend the banquet, they will decrease the price by $1.50 per person. What size group would maximize the profit for the hall this year and what would the maximum profit be?